MAE 1001: Python “Power Chords”

Introduction to Mechanical and Aerospace Engineering

Course website: https://gwu-mae1001.github.io/fall2026/

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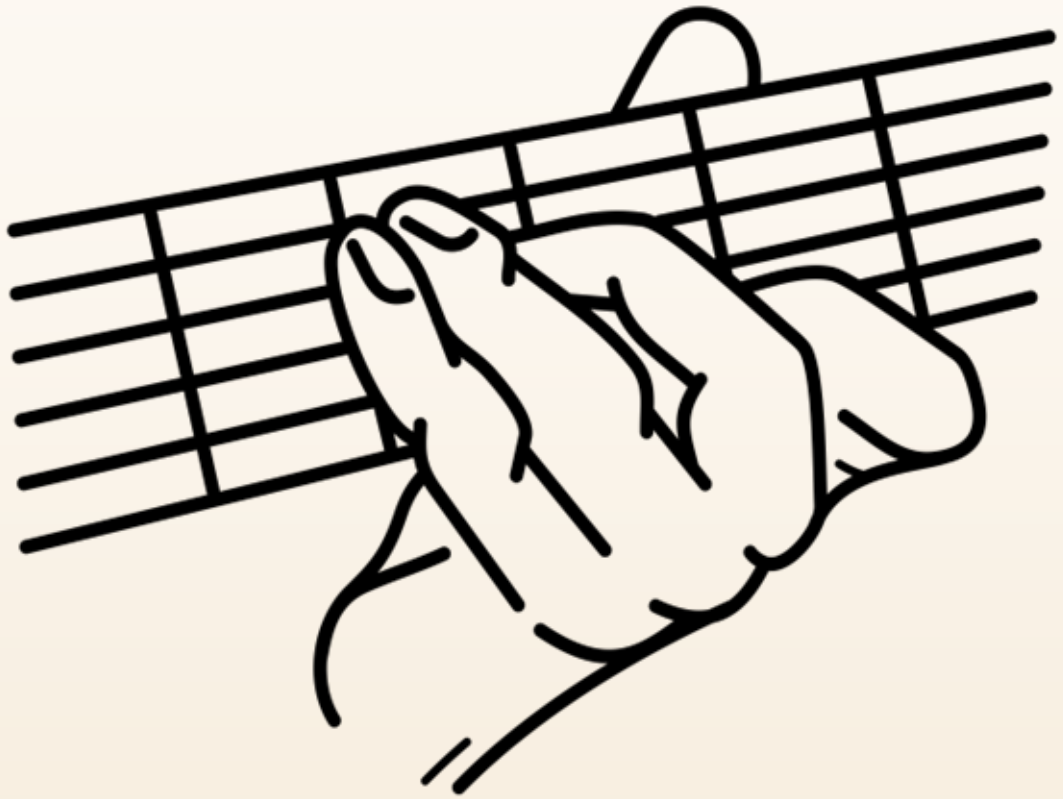
Fall 2026

Photo: Kartik Bulusu



"Study hard what interests you the most in the most undisciplined, irreverent and original manner possible."

— Richard Feynmann



What is a Power Chord?

- A power chord consists of just two notes: the root and the fifth.
- Commonly used in rock music for its strong, clear sound.

Representing Power Chords in Python

- Musical intervals can be modeled with arrays/lists, making it easy to calculate the root and fifth for any note.

```
import numpy as np
import sounddevice as sd

def play_power_chord(root_freq, duration=1.0, fs=44100):
    fifth_freq = root_freq * 2**(7/12)
    t = np.linspace(0, duration, int(fs*duration), endpoint=False)
    tone = np.sin(2*np.pi*root_freq*t) + np.sin(2*np.pi*fifth_freq*t)
    tone = tone / np.max(np.abs(tone))  # Normalize
    sd.play(tone, fs)
    sd.wait()

play_power_chord(82.41)
play_power_chord(110)
play_power_chord(146.83)
play_power_chord(196)
play_power_chord(246.94)
play_power_chord(329.63)
```

Skeleton of a Python program

Most programs follow the same top-to-bottom pattern.

```
1  """power_chords.py - play power chords"""

2  import numpy as np
   import sounddevice as sd

3  FS = 44100                    # sample rate (Hz)
   notes = {"E": 82.41, "A": 110.0, "D": 146.83}

4  def power_chord(root, duration=1.0):
       fifth = root * 2**(7/12)
       t = np.arange(int(FS * duration)) / FS
       tone = np.sin(2*np.pi*root*t) + np.sin(2*np.pi*fifth*t)
       return tone / np.max(np.abs(tone))

5  if __name__ == "__main__":
       for name, freq in notes.items():          6
           if freq < 100:                        7
               print(name, "- low chord")
           else:
               print(name, "- chord")
           sd.play(power_chord(freq), FS)        8
           sd.wait()
```

1. **Docstring & comments**
   Notes for people. Python skips them.
2. **Import modules & libraries**
   Borrow tools others wrote: NumPy, sounddevice.
3. **Declare variables**
   Give a value a name: a number, a dictionary of notes.
4. **Define a function (def)**
   Reusable block: inputs in, result returned.
5. **Main program**
   Runs when you run this file directly.
6. **Loop (for)**
   Repeat the indented block for every note.
7. **Decision (if / else)**
   Pick one path based on a condition.
8. **Call functions & methods**
   Use power_chord(); notes.items() is a method.

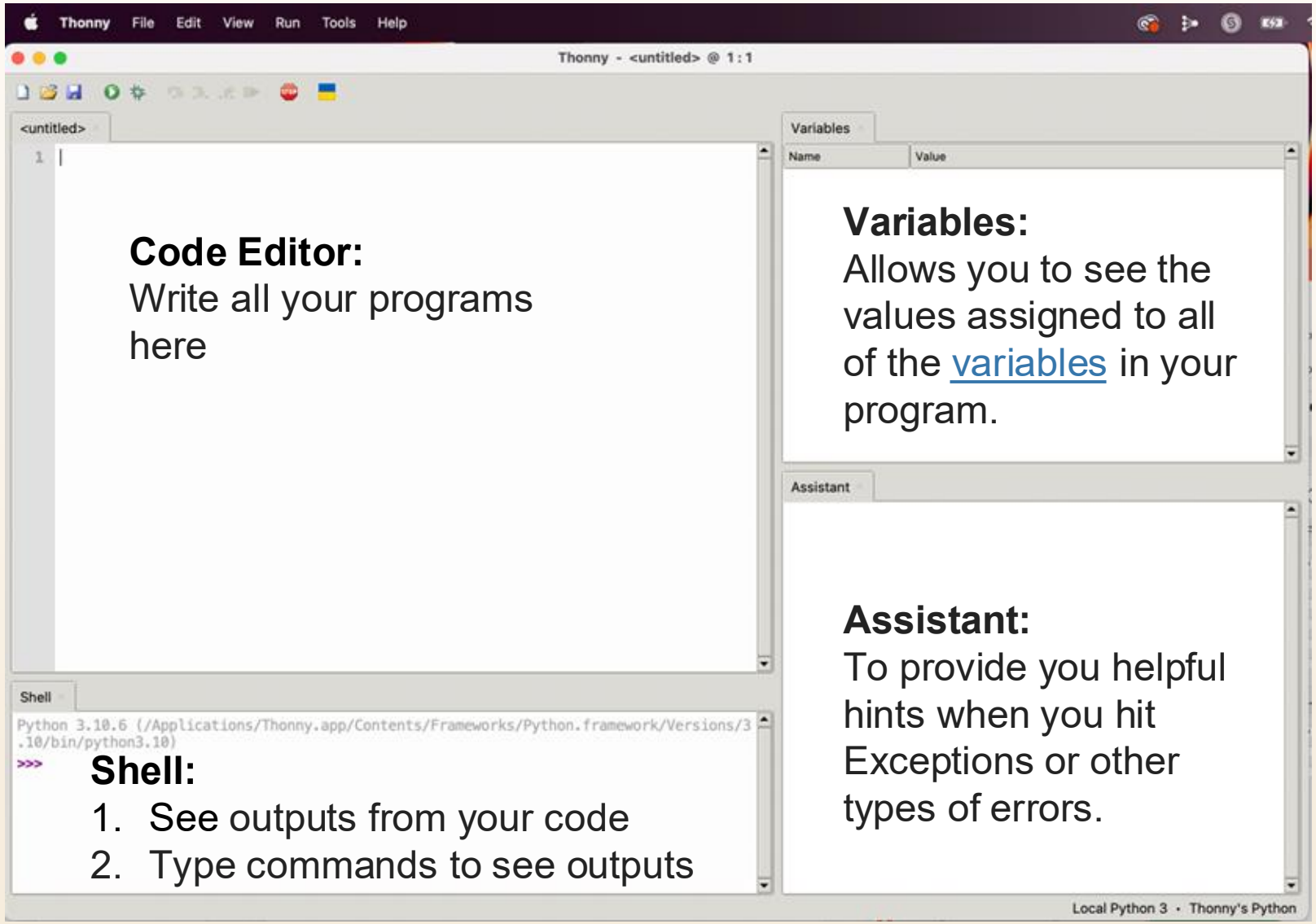
Sources:
Thonny (IDE): https://en.wikipedia.org/wiki/Thonny
Thonny: The Beginner-Friendly Python Editor: https://realpython.com/python-thonny/

Thonny-editor or integrated development environment (IDE)



Source:
https://www.raspberrypi.org/help/

What are the other options?



Code Editor:
Write all your programs here

Variables:
Allows you to see the values assigned to all of the variables in your program.

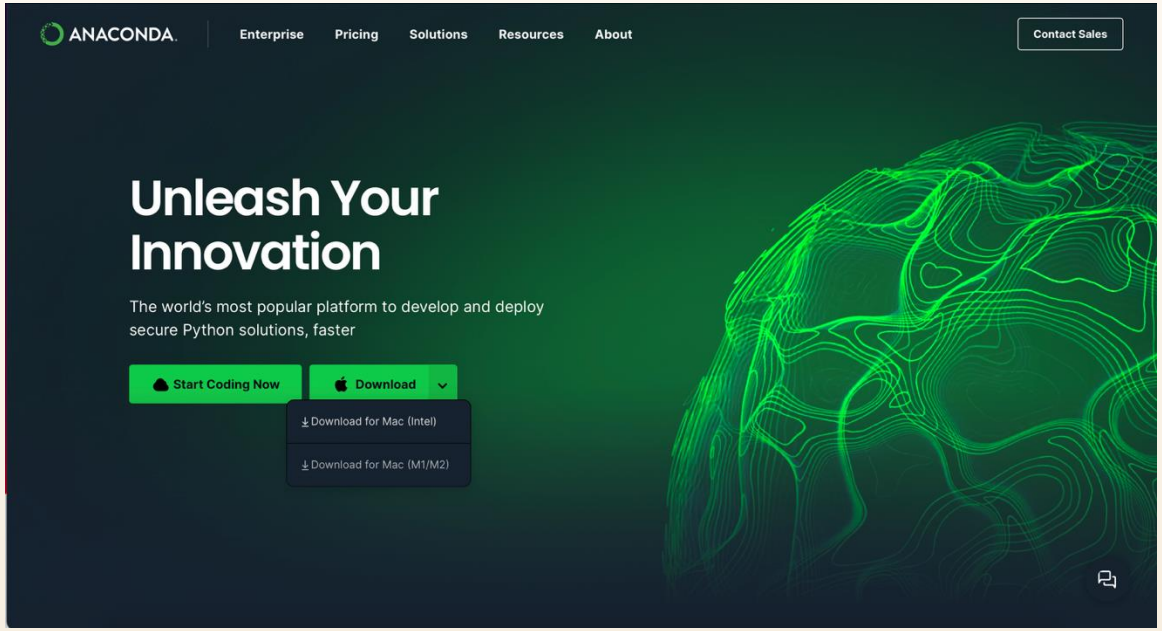
Assistant:
To provide you helpful hints when you hit Exceptions or other types of errors.

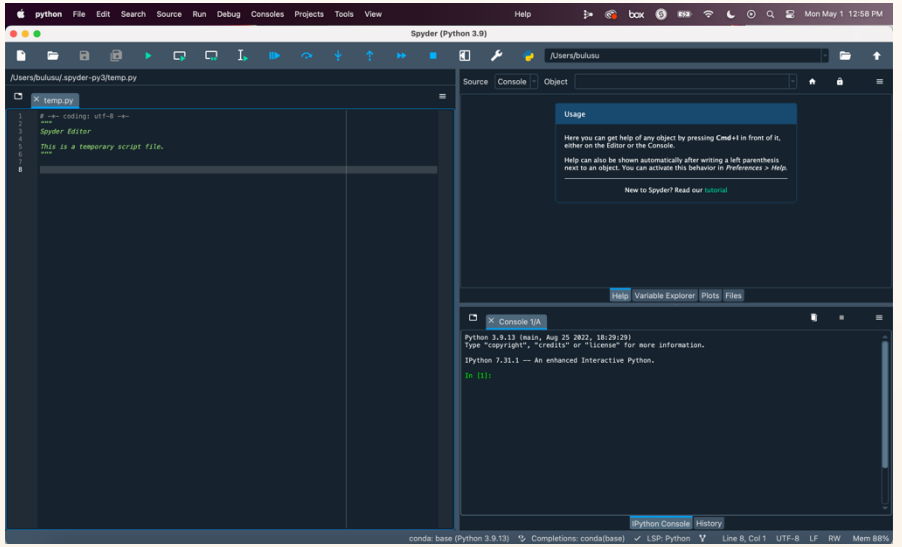
Shell:
1. See outputs from your code
2. Type commands to see outputs

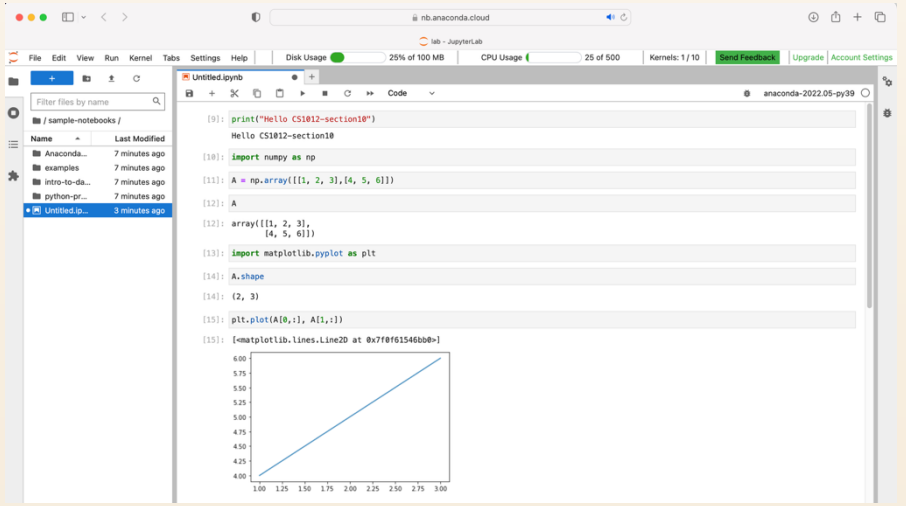
Anaconda python – IDEs:

- Spyder
- Jupyter Notebook
- Anaconda cloud

Source: Anaconda (IDE) https://www.anaconda.com

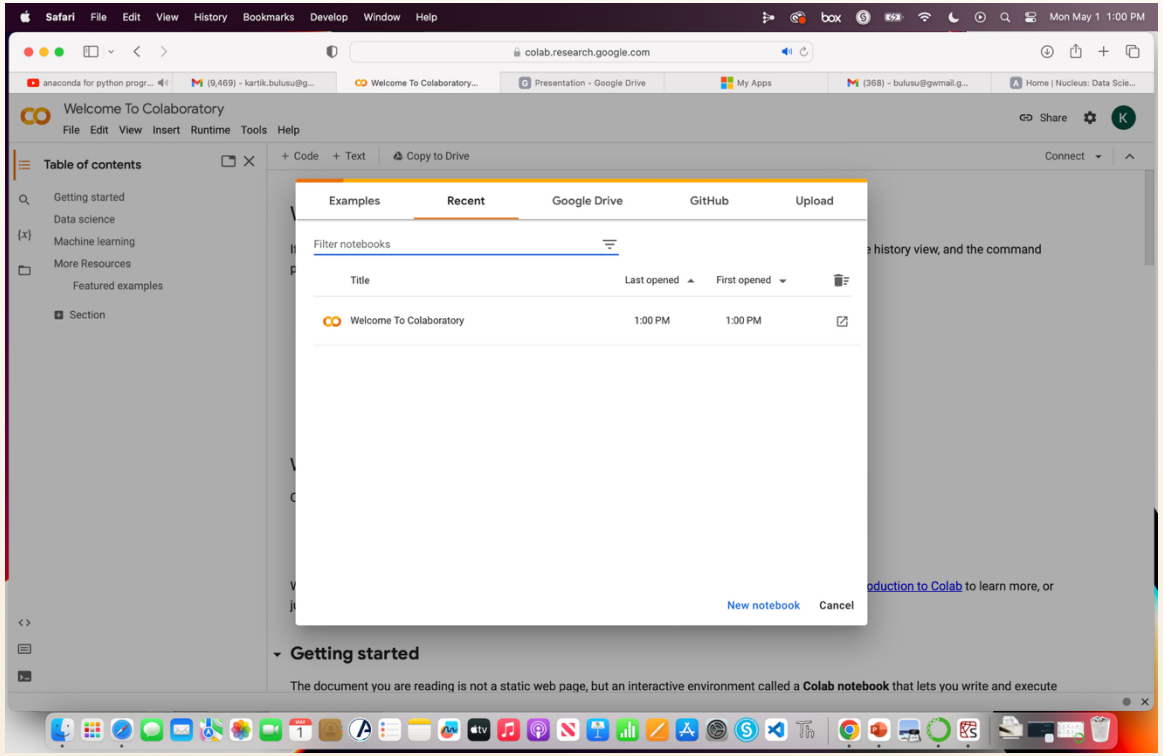






Google colab

Sources:
Google Colab https://colab.research.google.com/
Google Workspace Marketplace https://workspace.google.com/
Youtube video shorts https://youtube.com/shorts/7J3gU34j7Zk?feature=share





Colaboratory ("Colab" for short)

- Data analysis and machine learning tool that allows you to
  - combine executable Python code and
  - rich text along with charts, images, HTML, LaTeX and
  - more into a single document stored in Google Drive.



It connects to powerful Google Cloud Platform runtimes and enables you to easily share your work and collaborate with others.

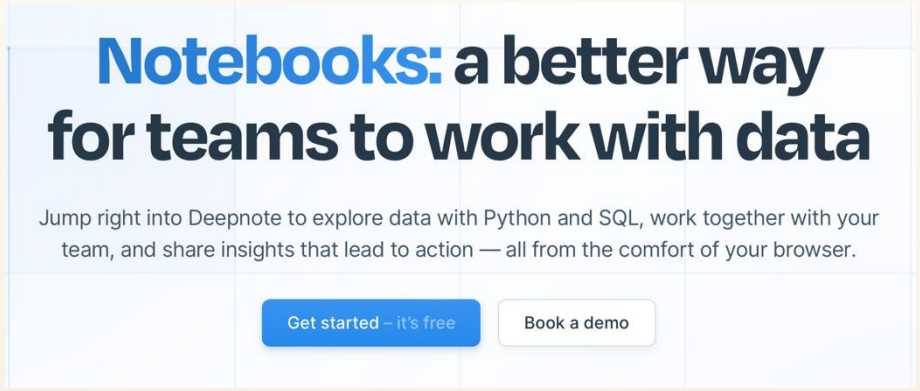
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Deepnote

Source:
https://deepnote.com







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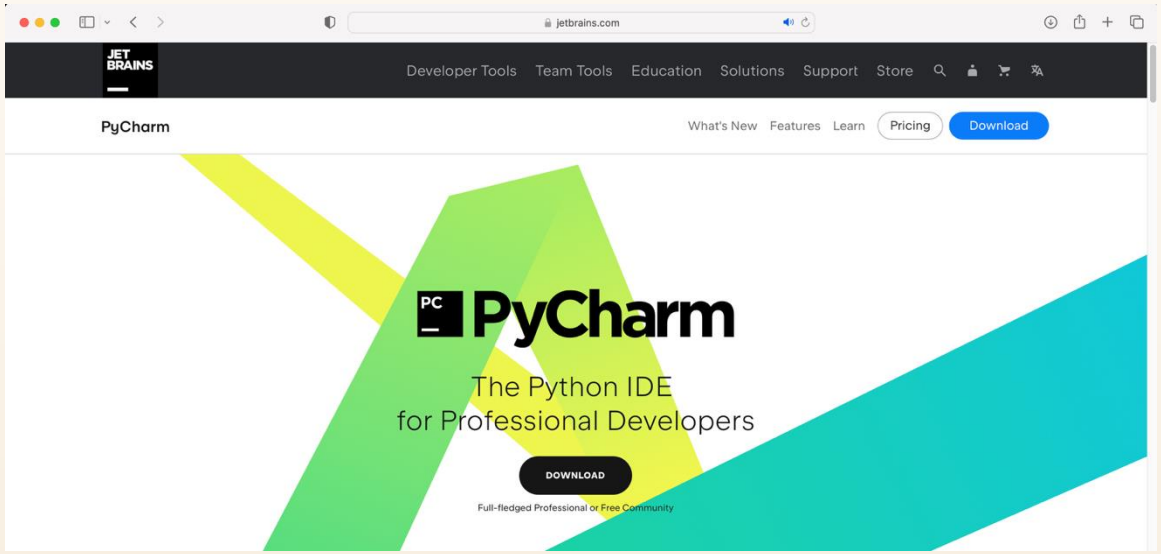
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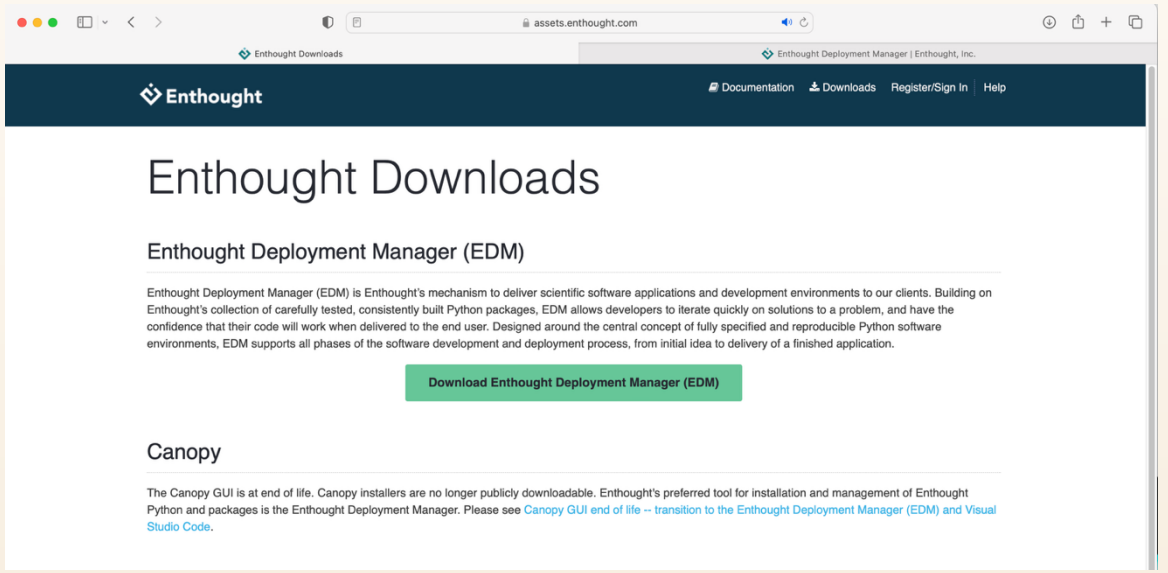
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More…

Sources:
Jetbrains-Pycharm: https://www.jetbrains.com/pycharm/
Enthought Deployment Manager (EDM): https://assets.enthought.com/downloads/





Three ways to run Python

Same language, three different homes for your code.

Python scripts



Write it all, run it all

Like recording a full song

Thonny · Spyder · PyCharm

Notebooks

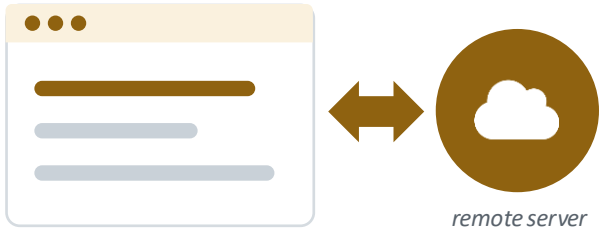


Run a cell, see the result

Like a jam session

Jupyter Notebook · JupyterLab

Cloud-based Python



Your browser, their computer

Like a borrowed studio

Colab · Deepnote · Anaconda Cloud

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How they differ

At a glance: more filled dots means a better fit.

	Python scripts	Notebooks	Cloud-based Python
Runs on	Your laptop	Your laptop	Remote server
File type	`.py`	`.ipynb`	`.ipynb`
Quick to start	●● ○	●● ○	●●●
Explore step by step	● ○○	●●●	●●●
Hardware access	●●●	●●●	● ○○
Share & collaborate	● ○○	●● ○	●●●
Works offline	✓	✓	✗

Advantages and disadvantages

What you gain, and what you give up.

Python scripts

GAINS

- Predictable runs
- Talks to hardware
- Easy to automate

TRADE-OFFS

- Re-run everything
- Plots pop out
- Clunky for exploring

Notebooks

GAINS

- Instant feedback
- Code + plots + notes
- Great for lab reports

TRADE-OFFS

- Out-of-order bugs
- Hard to reuse
- Needs local install

Cloud-based Python

GAINS

- Zero setup
- Free GPUs
- Live teamwork

TRADE-OFFS

- Needs internet
- No local hardware
- Sessions time out

Which one should you reach for?

Start from what you need, not from the tool.

IF YOU NEED…			REACH FOR…
Hardware	Automation	Reusable code	Python scripts
Exploring data	Quick plots	Lab reports	Notebooks
No install	GPU power	Teamwork	Cloud-based Python

Mix and match: explore in a notebook, then save the finished code as a .py script.